Jayesh Sangave

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EDUCATION

MIT World Peace University **2023 – 2027**
B.Tech in Computer Science Engineering *Pune, MH*

WORK EXPERIENCE

MIT Tech Team (MTT), ROBOCON **2023 – Present**
Robotics Control Developer | DD Robocon, 100+ national teams *Pune, MH*

Team Mentor *2026 – Present*
- Mentoring junior engineers in robotics, controls, embedded systems, ROS2, and competition robot development.

Team Captain & Manual Operator *2026*
- Led the team at DD Robocon 2026 across autonomous and manual robot development, testing, strategy, and operations, finishing **All India Rank 2**.
- Served as the manual operator during competition while overseeing both robots.

Head of Finance & Management *2025 – 2026*
- Led budgeting, procurement, sponsorships, and operations for the 2025–26 cycle; secured multiple sponsors.
- Planned and ran technical workshops and university events, handling both technical delivery and logistics.

Senior Engineer & Manual Operator *2025*
- Led end-to-end development of the jump robot (design to competition), built its core control and LiDAR auto-aim system, and operated it in matches.

Trainee Engineer *2024*
- Built ROS-based navigation for R2 (2024) and integrated STM32/ESP32 nodes over CAN/UART for manual robots.

ACHIEVEMENTS AND PROJECTS

Invited Student Speaker: World Technology Summit 2025–26
- Presented technologies developed by MIT Tech Team to international delegates, representing the team's robotics work on a global platform.

DD Robocon [2023] [2024] [2025] [2026]
- **AIR 2 (2026)** and **AIR 3 (2024)** among 100+ national teams; built a jump-and-dunk robot for the 2025 season.

Autonomous Navigation: R2 Robot, Robocon 2024 *2024*
- Optimized ROS-based navigation on the R2 competition robot, cutting full-field traversal time from **15 s to 7 s** (**53% faster**).

Dunking Robot: Robocon 2025 *Jan 2025 – Mar 2025*
- Designed and built India's only dunking robot: a **25 kg robot that jumps 30 cm** using an elastic-based energy storage system.

Real-Time Auto-Aim System: Robocon 2025 *2025*
- Engineered a real-time auto-aim system for closed-loop self-alignment and dynamic shooting-power adjustment.

SKILLS

Robotics & Autonomous Systems
ROS 2, Nav2, SLAM, LiDAR Localization, ICP, KISS-ICP, Motion Planning, Sensor Fusion, Autonomous Navigation, Gazebo

Computer Vision
OpenCV, Object Detection, Image Processing, Visual Odometry, Feature Tracking

Embedded & Controls
STM32, TM4C, ESP32, Embedded C, CAN, UART, SPI, I2C, PWM, BLDC Motors, Motor Control, Odometry, Real-Time Systems

Programming & Tools
C, C++, Python, Linux, Git, Docker, Ansible